

ECP-2400 EMERGENCY COOLING PUMP SYSTEM - TECHNICAL MANUAL

Document ID: ECP-2400-OM-REV4 | Department: Industrial Fluid Dynamics

1. GENERAL SPECIFICATIONS AND PERFORMANCE

The Model ECP-2400 is an industrial emergency cooling pump skid designed for critical heat exchange and thermal plant containment operations.

1.1 Operational Ratings

- Nominal operating speed: 2400 RPM (direct drive synchronous coupling).
- Maximum allowable operating pressure: 12.5 bar (test pressure 18.75 bar).
- Rated liquid flow rate: 350 L/min at nominal speed under standard ambient conditions.
- Maximum suction lift capability: 6.2 meters at sea level (101.3 kPa barometric pressure).
- Permissible particulate size in pumped media: maximum 0.5 mm non-abrasive suspended solids.

1.2 Skid Dimensions and Physical Characteristics

- Total skid assembly dry weight: 420 kg (operating weight 465 kg with flooded casing).
- Skid footprint dimensions: 1200 mm length by 800 mm width (overall height 950 mm).
- Suction flange diameter: DN80 PN16; Discharge flange diameter: DN50 PN16.

2. LUBRICATION, SEALS, AND MECHANICAL WEAR LIMITS

2.1 Lubricant Specifications

- Specified lubricant for bearing housings: ISO VG 46 synthetic gear and bearing oil.
- Oil reservoir capacity: 1.8 liters per bearing housing.
- Oil change interval: 2000 operating hours or annually, whichever occurs first.

2.2 Mechanical Seal and Alignment Inspection

- Primary mechanical seal assembly inspection interval: conduct inspection every 500 operating hours.
- Check seal faces for scorch marks, carbon deposit buildup, or elastomer deterioration.
- Shaft alignment checks: conduct laser or dial indicator alignment checks every 3 months or immediately following any major maintenance intervention or motor relocation.

2.3 Fastener Torques and Impeller Wear Rings

- Casing flange M16 assembly bolts: tighten sequentially in a cross-star pattern to 85 Nm.
- Impeller wear ring replacement schedule: inspect radial clearance annually. Wear rings must be replaced whenever the measured radial clearance exceeds 0.8 mm (nominal clearance is 0.25 to 0.35 mm).

3. VIBRATION, THERMAL THRESHOLDS, ELECTRICAL SUPPLY & AIR BLEEDING

3.1 Vibration and Temperature Safety Thresholds

- Normal baseline overall vibration: below 2.2 mm/s RMS velocity.
- Cavitation warning threshold: vibration levels exceeding 4.5 mm/s RMS indicate severe cavitation; operators must immediately verify suction net positive suction head (NPSH) and throttle discharge.
- Emergency shutdown bearing temperature: 95 C (alarm warning trips at 85 C; automatic trip at 95 C).

3.2 Electrical Power Supply Requirements

- Power supply: 3-phase 400V AC at 50 Hz (tolerance +/- 5%).
- Motor power rating: 15 kW induction motor, rated full-load current 28.5 Amperes.

3.3 Mandatory Pre-Startup Air Bleeding Procedure

Before energizing the pump motor, all trapped air must be completely purged:

- Step 1: Open manual air vent valve V-2 located at the high point of the pump discharge casing.
- Step 2: Slowly crack open the suction line isolation valve to prime the impeller eye with fluid.
- Step 3: Allow liquid to displace trapped air until a solid, bubble-free fluid stream emerges from V-2.
- Step 4: Securely close manual vent valve V-2 and verify zero gland leakage prior to startup.